Recurrent Neural Networks (RNN)

Lab Objectives:

By the end of this lab, students will be able to:

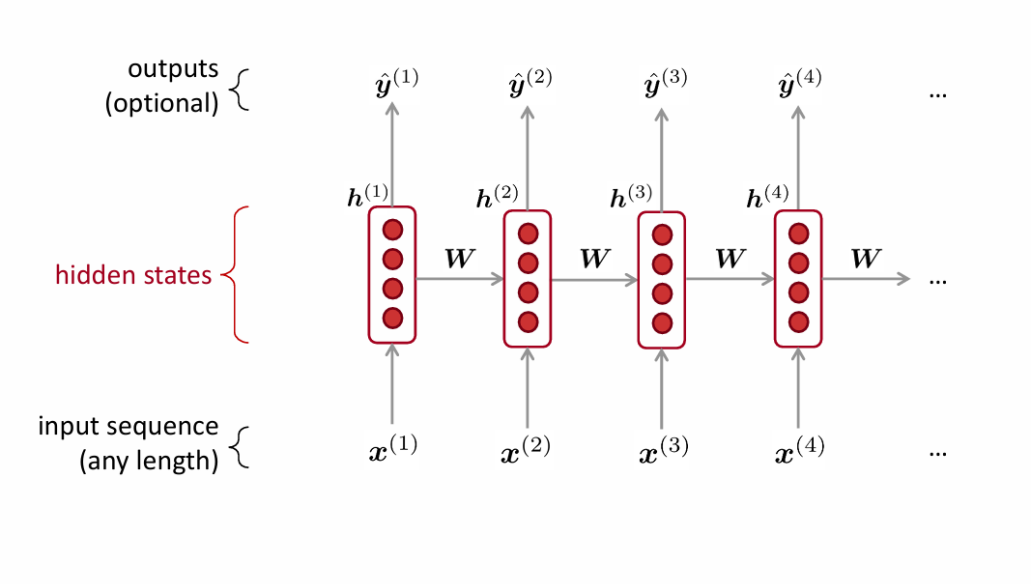
- Understand the implementation of RNN
- Create RNN in PyTorch
- Train and Evaluate RNN

1. What is RNN?

A recurrent neural network or RNN is a deep neural network trained on sequential or time series data to create a machine learning (ML) model that can make sequential predictions or conclusions based on sequential inputs.

2. How RNNs Work?

Like traditional neural networks, such as feedforward neural networks and convolutional neural networks (CNNs), recurrent neural networks use training data to learn. They are distinguished by their "memory" as they take information from prior inputs to influence the current input and output. While traditional deep learning networks assume that inputs and outputs are independent of each other, the output of recurrent neural networks depend on the prior elements within the sequence.



RNN tracks the context by maintaining a hidden state at each time step. A feedback loop is created by passing the hidden state from one-time step to the next. The hidden state acts as a memory that stores information about previous inputs. At each time step, the RNN processes the current input (for example, a word in a sentence) along

with the hidden state from the previous time step. This allows the RNN to "remember" previous data points and use that information to influence the current output.

Another distinguishing characteristic of recurrent networks is that they share parameters across each layer of the network. While feedforward networks have different weights across each node, recurrent neural networks share the same weight parameter within each layer of the network.

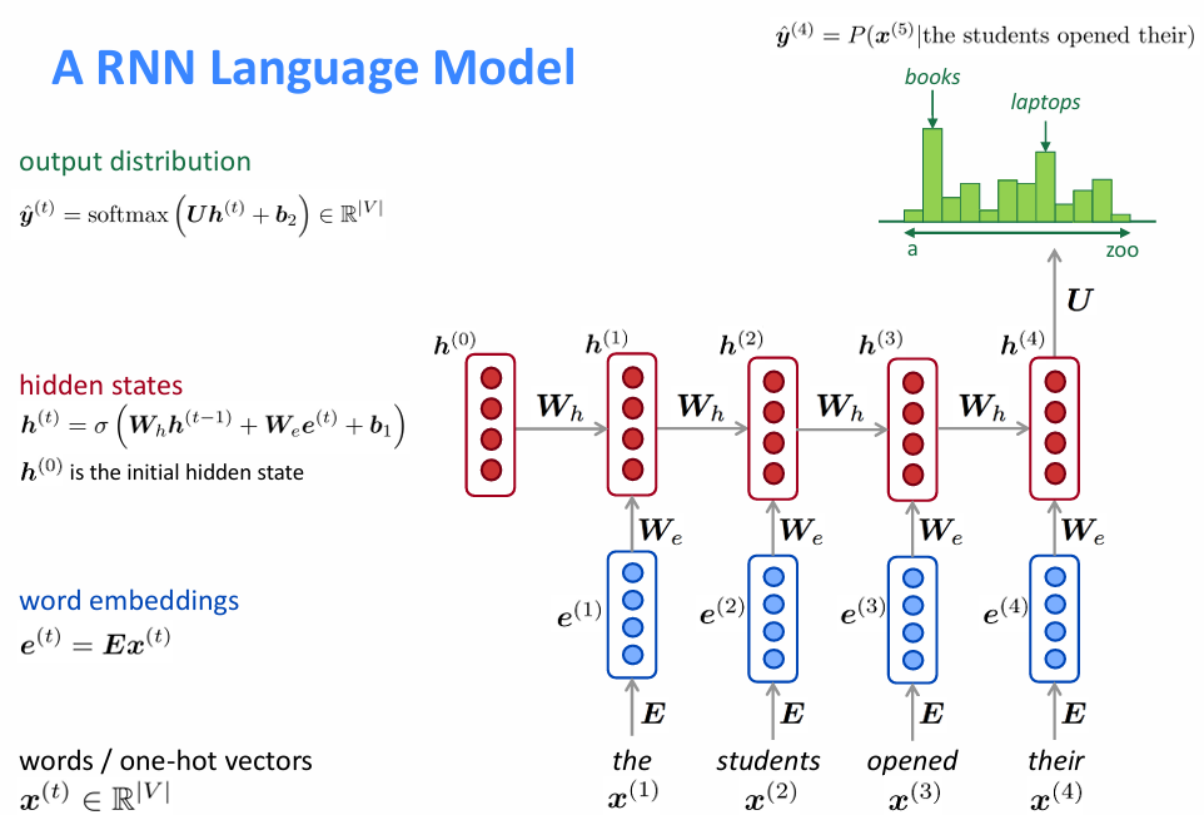
Recurrent neural networks use forward propagation and backpropagation through time (BPTT) algorithms to determine the gradients (or derivatives), which is slightly different from traditional backpropagation as it is specific to sequence data. The principles of BPTT are the same as traditional backpropagation, where the model trains itself by calculating errors from its output layer to its input layer. These calculations allow us to adjust and fit the parameters of the model appropriately. BPTT differs from the traditional approach in that BPTT sums errors at each time step whereas feedforward networks do not need to sum errors as they do not share parameters across each layer.

3. **Activation Functions:**

An activation function is a mathematical function applied to the output of each layer of neurons in the network to introduce nonlinearity and allow the network to learn more complex patterns in the data. In RNNs, activation functions are applied at each time step to the hidden states, controlling how the network updates its internal memory (hidden state) based on current input and past hidden states.

4. **Types of RNNs:**

- Standard RNNs
- Bidirectional recurrent neural networks (BRRNs)
- Long short-term memory (LSTM)
- Gated recurrent units (GNUs)'
- Encoder-decoder RNN



Classifying Names with character-level-RNN:

- **Preparing Setup:**

```
import torch

# Check if CUDA is available
device = torch.device('cpu')
if torch.cuda.is_available():
    device = torch.device('cuda')

torch.set_default_device(device)
print(f"Using device = {torch.get_default_device()}")
```

- **Preparing the Data:**

  Download the data from here and extract it to the current directory included in the data/names directory are 18 text files named as [Language].txt. Each file contains a bunch of names, one name per line, mostly romanized (but we still need to convert from Unicode to ASCII).

The first step is to define and clean our data. Initially, we need to convert Unicode to plain ASCII to limit the RNN input layers. This is accomplished by converting Unicode strings to ASCII and allowing only a small set of allowed characters.

```
import string
import unicodedata

# We can use "_" to represent an out-of-vocabulary character, that is, any character we are not handling in our model
allowed_characters = string.ascii_letters + " .,;'" + "_"
n_letters = len(allowed_characters)

# Turn a Unicode string to plain ASCII, thanks to https://stackoverflow.com/a/518232/2809427
def unicodeToAscii(s):
    return ''.join(
        c for c in unicodedata.normalize('NFD', s)
        if unicodedata.category(c) != 'Mn'
        and c in allowed_characters
    )
```

Here's an example of converting a unicode alphabet name to plain ASCII. This simplifies the input layer

```
print (f"converting 'Ślusàrski' to {unicodeToAscii('Ślusàrski')}")
```

- **Turning Names into Tensors:**

Now that we have all the names organized, we need to turn them into Tensors to make any use of them. To represent a single letter, we use a "one-hot vector" of size <1 x n_letters>. A one-hot vector is filled with 0s except for a 1 at index of the current letter, e.g. "b" = <0 1 0 0 0 ...>. To make a word we join a bunch of those into a 2D matrix <line_length x 1 x n_letters>.

That extra 1 dimension is because PyTorch assumes everything is in batches - we're just using a batch size of 1 here.

```
# Find letter index from all_letters, e.g. "a" = 0
def letterToIndex(letter):
    # return our out-of-vocabulary character if we encounter a letter unknown to our model
    if letter not in allowed_characters:
        return allowed_characters.find("_")
    else:
        return allowed_characters.find(letter)

# Turn a line into a <line_length x 1 x n_letters>,
# or an array of one-hot letter vectors
def lineToTensor(line):
    tensor = torch.zeros(len(line), 1, n_letters)
    for li, letter in enumerate(line):
        tensor[li][0][letterToIndex(letter)] = 1
    return tensor
```

Here are some examples of how to use lineToTensor() for a single and multiple character string.

```
print (f"The letter 'a' becomes {lineToTensor('a')}") #notice that the first position in the tensor = 1
print (f"The name 'Ahn' becomes {lineToTensor('Ahn')}") #notice 'A' sets the 27th index to 1
```

Next, we need to combine all our examples into a dataset so we can train, test and validate our models. For this, we will use the Dataset and DataLoader classes to hold our dataset. Each Dataset needs to implement three functions: __init__, __len__, and __getitem__.

```
from io import open
import glob
import os
import time

import torch
from torch.utils.data import Dataset

class NamesDataset(Dataset):

    def __init__(self, data_dir):
        self.data_dir = data_dir #for provenance of the dataset
        self.load_time = time.localtime #for provenance of the dataset
        labels_set = set() #set of all classes

        self.data = []
        self.data_tensors = []
        self.labels = []
        self.labels_tensors = []

        #read all the ``.txt`` files in the specified directory
        text_files = glob.glob(os.path.join(data_dir, '*.txt'))
        for filename in text_files:
            label = os.path.splitext(os.path.basename(filename))[0]
            labels_set.add(label)
            lines = open(filename, encoding='utf-8').read().strip().split('\n')
            for name in lines:
                self.data.append(name)
                self.data_tensors.append(lineToTensor(name))
                self.labels.append(label)

        #Cache the tensor representation of the labels
        self.labels_uniq = list(labels_set)
        for idx in range(len(self.labels)):
            temp_tensor = torch.tensor([self.labels_uniq.index(self.labels[idx])], dtype=torch.long)
            self.labels_tensors.append(temp_tensor)

    def __len__(self):
        return len(self.data)

    def __getitem__(self, idx):
        data_item = self.data[idx]
        data_label = self.labels[idx]
        data_tensor = self.data_tensors[idx]
        label_tensor = self.labels_tensors[idx]

        return label_tensor, data_tensor, data_label, data_item
```

Here we can load our example data into the NamesDataset.

```
alldata = NamesDataset("data/names")
print(f"loaded {len(alldata)} items of data")
print(f"example = {alldata[0]}")
```

Using the dataset object allows us to easily split the data into train and test sets. Here we create a 85/15 split.

```
train_set, test_set = torch.utils.data.random_split(alldata, [.85, .15], generator=torch.Generator(device=device).manual_seed(2024))

print(f"train examples = {len(train_set)}, validation examples = {len(test_set)}")
```

- **Creating the Network:**

This CharRNN class implements an RNN with three components. First, we use the nn.RNN implementation. Next, we define a layer that maps the RNN hidden layers to our output. And finally, we apply a softmax function.

```
import torch.nn as nn
import torch.nn.functional as F

class CharRNN(nn.Module):
    def __init__(self, input_size, hidden_size, output_size):
        super(CharRNN, self).__init__()

        self.rnn = nn.RNN(input_size, hidden_size)
        self.h2o = nn.Linear(hidden_size, output_size)
        self.softmax = nn.LogSoftmax(dim=1)

    def forward(self, line_tensor):
        rnn_out, hidden = self.rnn(line_tensor)
        output = self.h2o(hidden[0])
        output = self.softmax(output)

        return output
```

We can then create an RNN with 58 input nodes, 128 hidden nodes, and 18 outputs:

```
n_hidden = 128
rnn = CharRNN(n_letters, n_hidden, len(alldata.labels_uniq))
print(rnn)
```

After that we can pass our Tensor to the RNN to obtain a predicted output. Subsequently, we use a helper function, label_from_output, to derive a text label for the class.

```
def label_from_output(output, output_labels):
    top_n, top_i = output.topk(1)
    label_i = top_i[0].item()
    return output_labels[label_i], label_i

input = lineToTensor('Albert')
output = rnn(input) #this is equivalent to ``output = rnn.forward(input)``
print(output)
print(label_from_output(output, alldata.labels_uniq))
```

- **Training the Network:**

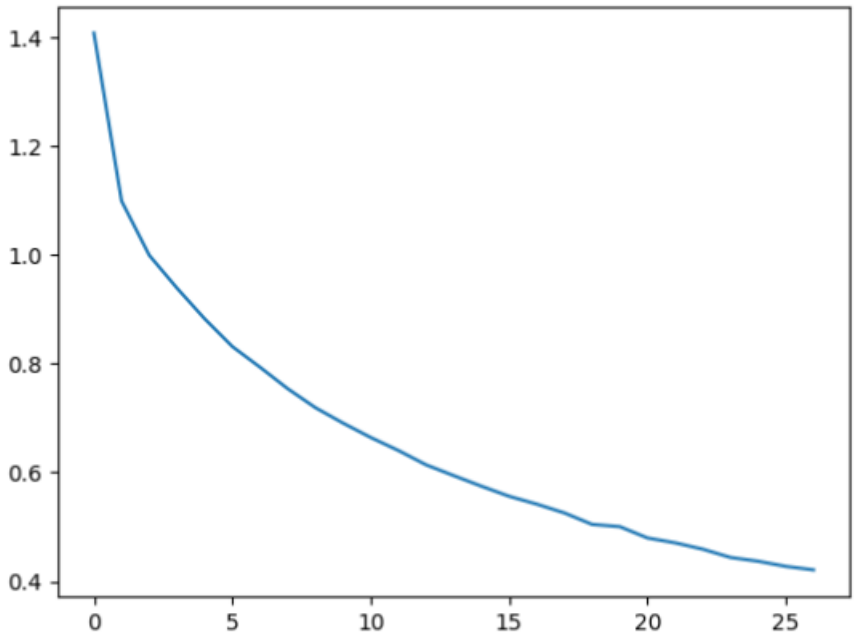
We do this by defining a train() function which trains the model on a given dataset using minibatches. RNNs RNNs are trained similarly to other networks; therefore, for completeness, we include a batched training method here. The loop (for i in batch) computes the losses for each of the items in the batch before adjusting the weights. This operation is repeated until the number of epochs is reached.

```
import random
import numpy as np

def train(rnn, training_data, n_epoch = 10, n_batch_size = 64, report_every = 50, learning_rate = 0.2, criterion = nn.NLLLoss()):
    """
    Learn on a batch of training_data for a specified number of iterations and reporting thresholds
    """
    # Keep track of losses for plotting
    current_loss = 0
    all_losses = []
    rnn.train()
    optimizer = torch.optim.SGD(rnn.parameters(), lr=learning_rate)

    start = time.time()
    print(f"training on data set with n = {len(training_data)}")

    for iter in range(1, n_epoch + 1):
        rnn.zero_grad() # clear the gradients

        # create some minibatches
        # we cannot use dataloaders because each of our names is a different length
        batches = list(range(len(training_data)))
        random.shuffle(batches)
        batches = np.array_split(batches, len(batches) //n_batch_size )

        for idx, batch in enumerate(batches):
            batch_loss = 0
            for i in batch: #for each example in this batch
                (label_tensor, text_tensor, label, text) = training_data[i]
                output = rnn.forward(text_tensor)
                loss = criterion(output, label_tensor)
                batch_loss += loss

            # optimize parameters
            batch_loss.backward()
            nn.utils.clip_grad_norm_(rnn.parameters(), 3)
            optimizer.step()
            optimizer.zero_grad()

            current_loss += batch_loss.item() / len(batch)

        all_losses.append(current_loss / len(batches) )
        if iter % report_every == 0:
            print(f"{iter} ({iter / n_epoch:.0%}): \t average batch loss = {all_losses[-1]}")
        current_loss = 0

    return all_losses
```

```
start = time.time()
all_losses = train(rnn, train_set, n_epoch=27, learning_rate=0.15, report_every=5)
end = time.time()
print(f"training took {end-start}s")
```

- **Plotting the Results:**

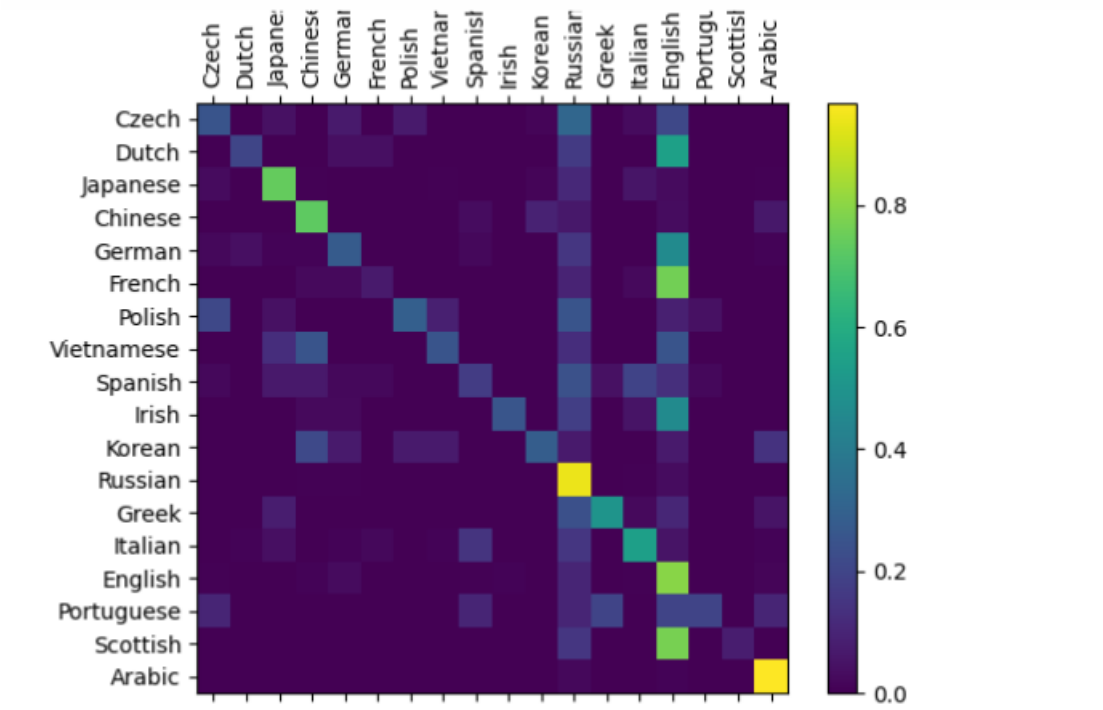
```
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker

plt.figure()
plt.plot(all_losses)
plt.show()
```



- **Evaluating the Results:**

  To see how well the network performs on different categories, we will create a confusion matrix, indicating for every actual language (rows) which language the network guesses (columns). To calculate the confusion matrix a bunch of samples are run through the network with evaluate(), which is the same as train() minus the backprop.

```python
def evaluate(rnn, testing_data, classes):
    confusion = torch.zeros(len(classes), len(classes))

    rnn.eval() #set to eval mode
    with torch.no_grad(): # do not record the gradients during eval phase
        for i in range(len(testing_data)):
            (label_tensor, text_tensor, label, text) = testing_data[i]
            output = rnn(text_tensor)
            guess, guess_i = label_from_output(output, classes)
            label_i = classes.index(label)
            confusion[label_i][guess_i] += 1

    # Normalize by dividing every row by its sum
    for i in range(len(classes)):
        denom = confusion[i].sum()
        if denom > 0:
            confusion[i] = confusion[i] / denom

    # Set up plot
    fig = plt.figure()
    ax = fig.add_subplot(111)
    cax = ax.matshow(confusion.cpu().numpy()) #numpy uses cpu here so we need to use a cpu version
    fig.colorbar(cax)

    # Set up axes
    ax.set_xticks(np.arange(len(classes)), labels=classes, rotation=90)
    ax.set_yticks(np.arange(len(classes)), labels=classes)

    # Force label at every tick
    ax.xaxis.set_major_locator(ticker.MultipleLocator(1))
    ax.yaxis.set_major_locator(ticker.MultipleLocator(1))

    # sphinx_gallery_thumbnail_number = 2
    plt.show()



evaluate(rnn, test_set, classes=alldata.labels_uniq)
```



Lab Exercise:

- Practice the above example.
- Adjust the hyperparameters to enhance performance, such as changing the number of epochs, batch size, and learning rate.
- Try the nn.LSTM and nn.GRU layers.
- Modify the size of the layers, such as increasing or decreasing the number of hidden nodes or adding additional linear layers.